

CFT-150 Recirculating Chiller

**For GE Medical Systems
(50 Hertz and 60 Hertz Units)**

NESLAB Manual P/N U00460
Rev. 03/02/99

Installation, Operation and Service Manual

NESLAB online

Product Service Information, Applications
Notes, MSDS Forms, e-mail.

Call our bulletin board system

(603) 427-2490

Contact us for free access software.

Voice Info: (800) 4-NESLAB



Comments on this manual can be sent to:

NESLAB@neslabinstruments.com

or visit our Web page at:

<http://www.neslabinc.com>

CFT-150 Recirculating Chiller

PREFACE

Compliance	2
Unpacking	2
After-sale Support	2

SECTION I

Safety

Warnings	3
Additional Warnings	4

SECTION II

General Information

Description	5
Specifications	5

SECTION III

Installation

Site	7
Electrical Requirements	8
Plumbing Requirements	8
Fluids	8
Filling Requirements	9

SECTION IV

Operation

Start Up	10
Digital Controller	11
Operator's Loop	12

SECTION V

Maintenance & Service

Setup Loop	13
Error Codes	14
Displaying Software Version Number	15
Cleaning	15
Pump Strainer	16
Leaks	16
Pressure Relief Valves	17
Algae	18
Pump Motor Lubrication	18
Hoses	18
Refrigeration Data	18
Spare Parts List	19
Factory Preset Values	19

SECTION VI

Troubleshooting

General	21
Starting	23
No Flow	24
Pump Noise	26
Low Flow and/or Pressure	28
Pressure Gauge Always Zero	29
Pressure Gauge Incorrect	30
Motor Operation	31
Wiring Diagrams	32

Preface

Compliance

Products tested and found to be in compliance with the requirements defined in the EMC standards defined by 89/336/EEC as well as Low Voltage Directive (LVD) 73/23/EEC can be identified by the CE label on the rear of the unit. The testing has demonstrated compliance with the following directives:

LVD, 73/23/EEC	Complies with UL 3101-1:93
EMC, 89/336/EEC	EN 55011, Class A Verification
	EN 50082-1:1992
	IEC 1000-4-2:1995
	IEC 1000-4-3:1994
	IEC 1000-4-4:1995

For any additional information refer to the Letter of Compliance that shipped with the unit (Declaration of Conformity).

Unpacking

Retain all cartons and packing material until the unit is operated and found to be in good condition. If the unit shows external or internal damage, or does not operate properly, contact the transportation company and file a damage claim. Under ICC regulations, this is your responsibility.

After-sale Support

NESLAB is committed to customer service both during and after the sale. If you have questions concerning the unit operation, contact our Sales Department. If your unit fails to operate properly, or if you have questions concerning spare parts or Service Contracts, contact our Service Department.

Before calling, please refer to the labels on the rear of the unit to obtain the following information:

- unit BOM number _____

- unit serial number _____

The unit's serial number label and pump identification label are located on the rear of the unit.

	NESLAB Instruments, Inc. Newington, NH 03801 U.S.A. (800)258-0830 / (603)436-9444
BOM#: 337104041505	
S/N: 199266046	
MANUFACTURED FOR GENERAL ELECTRIC MEDICAL SYSTEMS GE P/N 2222564-2 MODEL# CFT-150	
VOLTAGE=208-230 AMPS=13.7 HZ=60 PH=1	
R-134A CHG 4LBS HI=300PSI LO=150PSI PUMP TYPE(S): PD2 1/3HP UL FILE # E16755	

Section I Safety

Warnings



Warnings are posted throughout the manual. These warnings are designated by an exclamation mark inside an equilateral triangle and text highlighted in bold. Read and follow these important instructions. Failure to observe these instructions can result in permanent damage to the unit, significant property damage, or personal injury or death.

Make sure you read and understand all instructions and safety precautions listed in this manual before installing or operating your unit. If you have any questions concerning the operation of your unit or the information in this manual, please contact our Sales Department (see After-sale Support).

Never place the unit in a location where excessive heat, moisture, or corrosive materials are present.

The unit construction provides extra protection against the risk of electrical shock by grounding appropriate metal parts. The extra protection may not function unless the power cord is connected to a properly grounded outlet. It is the user's responsibility to assure a proper ground connection is provided.

Never connect the OUTLET or INLET fitting to your building water supply or any water pressure source.

Never use flammable or corrosive fluids with this unit. Distilled and deionized water may be aggressive and cause material corrosion. Please contact NESLAB before subjecting this unit to prolonged exposure to highly distilled or deionized water.

Do not use automotive antifreeze. Commercial antifreeze contains silicates that can damage the pump seals. Use of automotive antifreeze will void the manufacturer's warranty.

Do not replace reservoir plug with a non-vented type or damage to the tank may occur.

Additional Warnings

In addition to the specific warnings listed on the previous page the following general warnings apply to your unit:

Performance of installation, operation, or maintenance procedures other than those described in this manual may result in a hazardous situation and may void the manufacturer's warranty.

Transport the unit with care. Sudden jolts or drops can damage the refrigeration lines.

Observe all warning labels.

Never remove warning labels.

Never operate damaged or leaking equipment.

Never operate the unit without cooling fluid in the reservoir.

Always turn off the unit and disconnect the power cord from the power source before performing any service or maintenance procedures, or before moving the unit.

Always empty the reservoir before moving the unit.

Never operate equipment with damaged power cords.

Refer service and repairs to a qualified technician.

Section II General Information

Description

The CFT-150 Recirculating Chiller is designed to provide a continuous supply of cooling fluid at a constant temperature and volume.

The unit consists of an air-cooled refrigeration system, a sealable reservoir, recirculating pump, and a temperature controller. Units are designed to run on either 60 Hertz or 50 Hertz frequency. 60 Hertz units are designed to UL-2601 specifications.

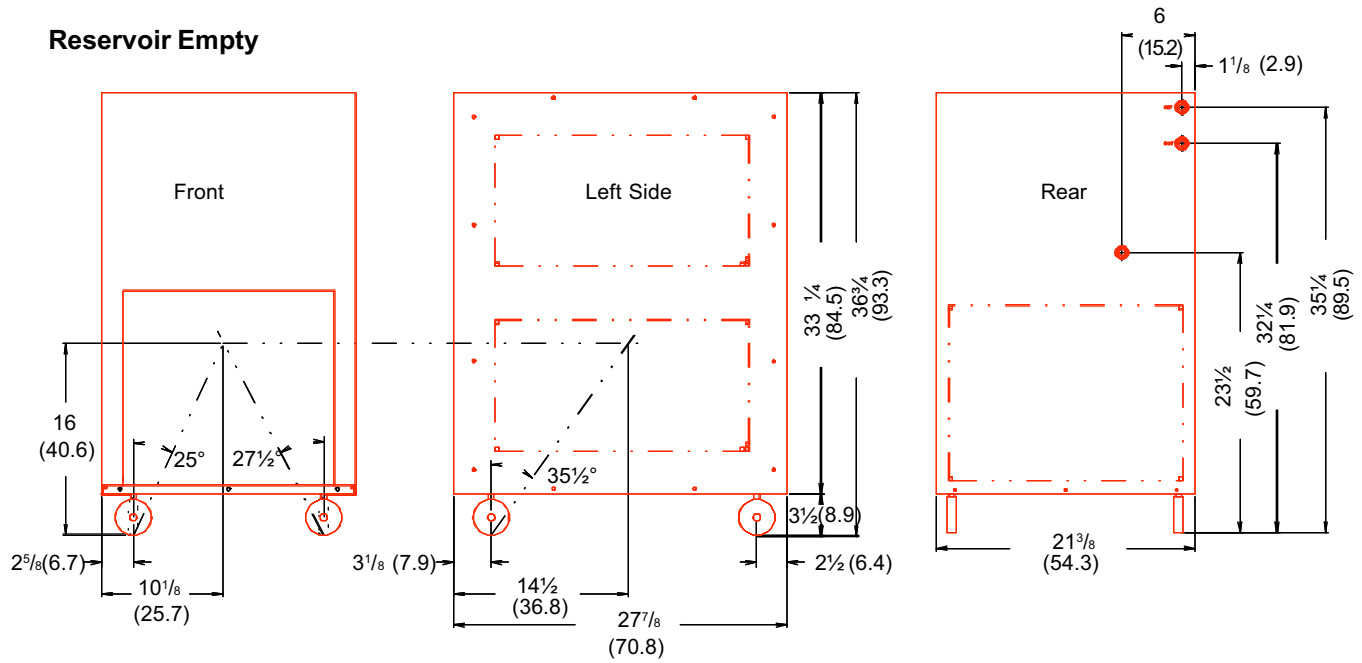
Specifications

Cooling Capacity¹ 60Hz 50Hz	4500 Watts 3735 Watts
Temperature Range	5°C to 30°C (41°F to 86°F)
Temperature Stability	±0.5°C (±1.1°F)
Pumping Capacity	Bar 4.0 3.4 2.7 2.0 1.3 0.7 PSI 60 50 30 30 20 10 50 Hz 60 Hz Flow 1 2 3 4 5 6 GPM 3.8 7.5 11.3 15.1 18.9 22.6 LPM
Reservoir Volume	5.0 Gallons 18.9 Liters
Refrigerant	R22
Air Intake Requirements	1050 Cubic feet per minute 29736 Liters per minute
Electrical Requirements	208-230V 60Hz 1Ø 13.7Amps or 208V 50 Hz 1Ø 12.6 Amps
Shipping Weight² (Dry)	310 Pounds 140 Kilograms

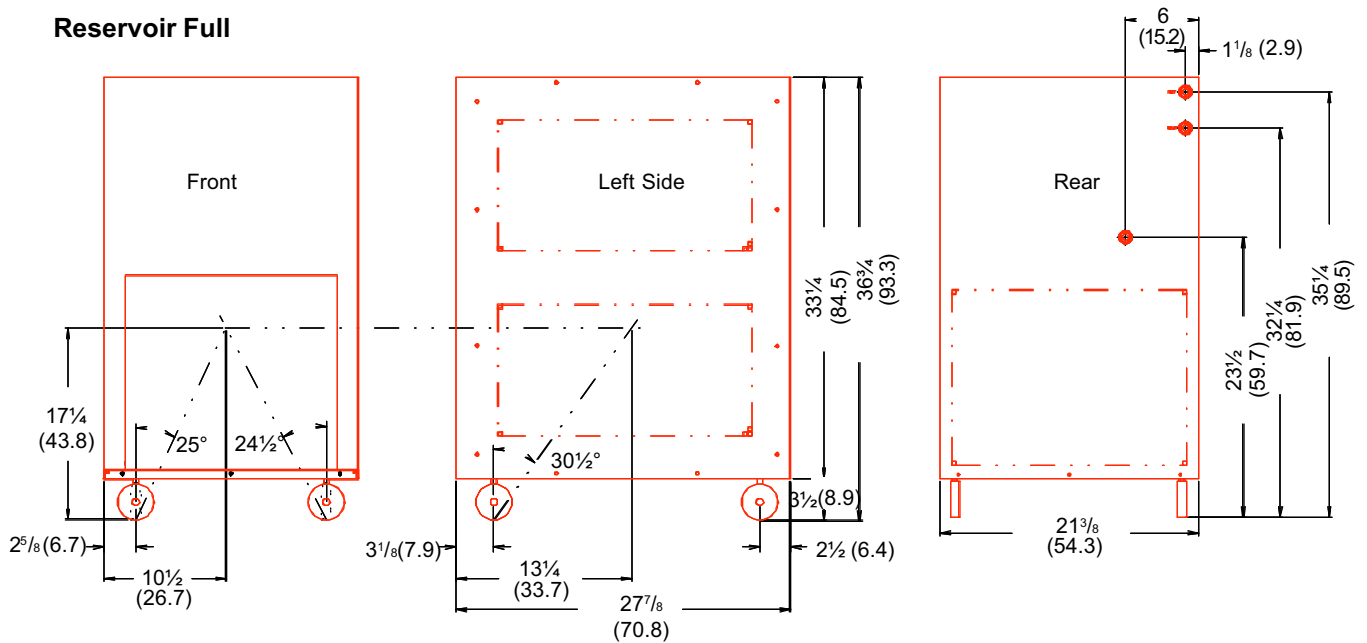
1. Circulating water at 20°C (68°F), at 20°C (68°F) ambient. Cooling capacity will vary depending on fluid temperature, ambient temperature, and cooling fluid.
2. Approximate.

Unit Dimensions/ Center of Gravity

Reservoir Empty



Reservoir Full



Dimensions are to the nearest $\frac{1}{8}$ inch (0.1 centimeter). Add approximately 2 inches (5 centimeters) to the depth to include the plumbing connections.

Section III Installation

Site

The unit should be located in a clean environment where ambient temperatures are inside the range of 13°C to 35°C (55°F to 95°F).

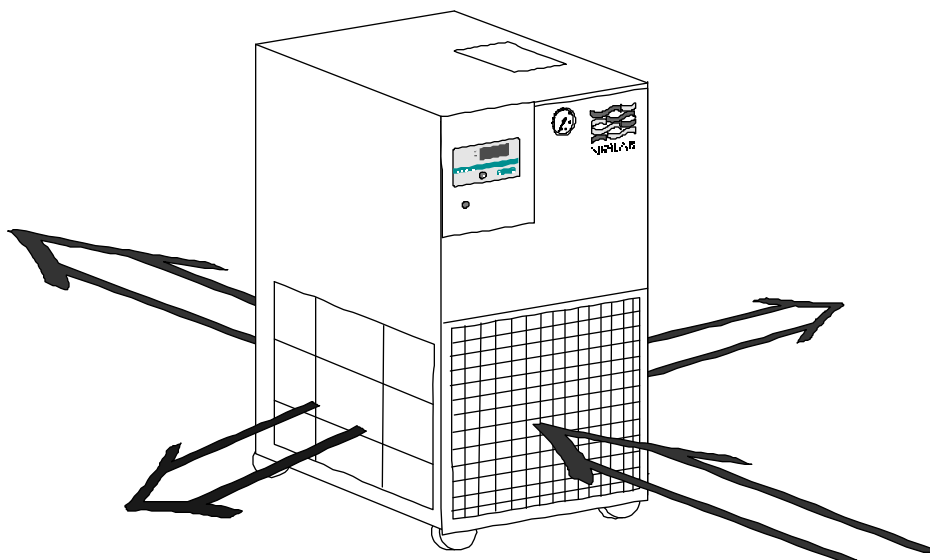


Never place the unit in a location where excessive heat, moisture, or corrosive materials are present.

The unit has an air-cooled refrigeration system. Air is drawn through the front of the unit and discharged through rear and side panels. The unit must be positioned so the intake and discharge are not impeded. A minimum clearance of 18 inches (45 centimeters) on all vented sides is necessary for adequate ventilation. Inadequate ventilation will cause a reduction in cooling capacity and, in extreme cases, compressor failure.

Excessively dusty areas should be avoided and a periodic cleaning schedule should be instituted (see Section V, Maintenance and Cleaning).

The unit will retain its full rated capacity in ambient temperatures up to approximately 24°C (75°F). Above 24°C (75°F), reduce the cooling capacity 1% for every 0.5°C (1°F) above 24°C (75°F), up to a maximum ambient temperature of 35°C (95°F).



Direction of Air Flow

Electrical Requirements



The unit construction provides extra protection against the risk of electrical shock by grounding appropriate metal parts. The extra protection may not function unless the power cord is connected to a properly grounded outlet. It is the user's responsibility to assure a proper ground connection is provided.

Refer to the serial number label on the rear of the unit for the specific electrical requirements of your unit.

Plumbing Requirements

The plumbing connections are located on the rear of the unit and are labelled OUTLET and INLET. The connections are ½ inch Female Pipe Thread.

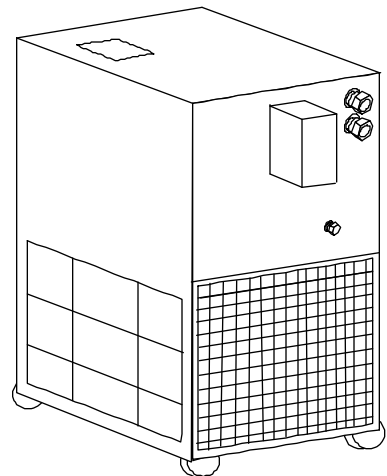
Remove the plastic protective plugs from both plumbing connections. Install the supplied ½" by ½" hose barb fittings to these connections. Extra/replacement barb fittings can be ordered from NESLAB, see Spare Parts List on page 22.

Connect the OUTLET fitting to the hose feeding the inlet of your application. Connect the INLET fitting to the hose from the outlet of your application. Clamp all connections.



Never connect the fittings to your building water supply or any water pressure source.

It is important to keep the distance between the unit and the instrument being cooled as short as possible. Tubing should be straight and without bends. If diameter reductions must be made, they should be made at the inlet and outlet of your application, not at the chiller.



Fluids

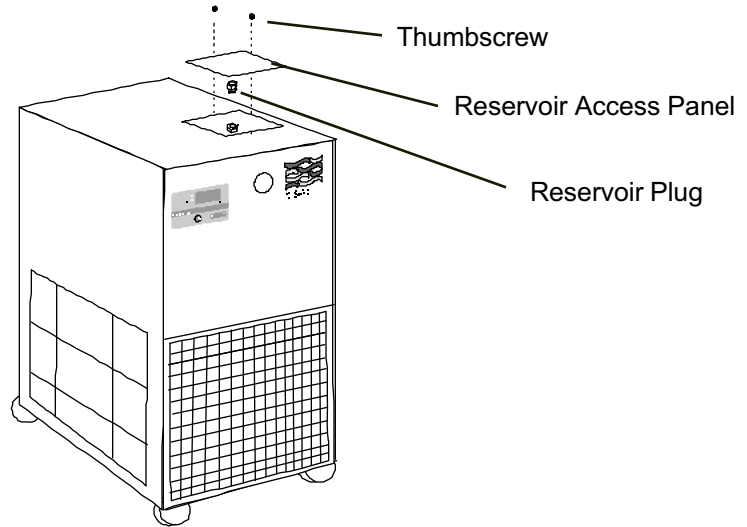


Never use flammable or corrosive fluids with this unit. Do not use automotive antifreeze. Commercial antifreeze contains silicates that can damage the pump seals. Use of automotive antifreeze will void the manufacturer's warranty.

NESLAB recommends using the antifreeze coolant provided by GE Medical Systems. See Spare Parts List on page 22.

Filling Requirements

Remove the reservoir access panel by unscrewing the thumbscrews. Locate the reservoir plug (square nut). Remove the plug and fill the reservoir with clean cooling fluid, following the special considerations outlined in the following paragraphs.



Reservoir filling location

Fill the reservoir to the bottom of the fill hole threads. Since the reservoir capacity is small compared to your application, have extra cooling fluid on hand to keep the system topped off when external circulation is started.

If substantial lengths of cooling lines are required, they should be pre-filled with coolant before connecting them to the unit.

NOTE: The tank in your unit has a vent which relieves pressure built up from thermal expansion of water. The vent is located on the reservoir plug. It activates when tank pressure reaches 3 - 5 psi.



Do not replace reservoir plug with an non-vented type or damage to the tank may occur.

Section IV Operation

Start Up

Before starting the unit, double check all electrical and plumbing connections and make sure the circulating system (the CFT, the instrument being cooled, and the tubing that connects them) has been properly filled with cooling fluid. Place the circuit breaker located on the rear of the unit to the up position. To start the unit, place the POWER Switch to the ON position. The refrigeration system and the recirculation pump will start. The pump operating pressure will be displayed on the RECIRCULATING PRESSURE gauge.

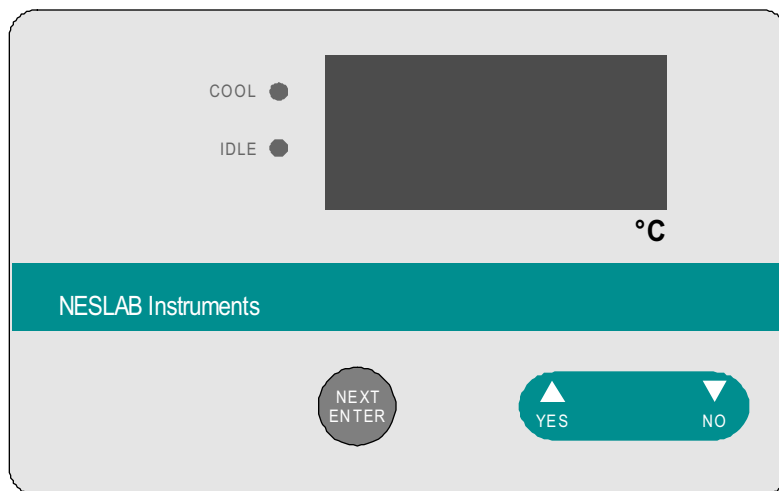
Ensure the setpoint is 20°C. See page 12 for the procedure.

The COOL LED indicates the status of the refrigeration system. It illuminates to indicate the refrigeration system is removing heat from the cooling fluid. As the operating temperature approaches the temperature setpoint, the LED will flash.

The IDLE LED indicates the unit is in a hot-gas-bypass mode of operation. As the operating temperature approaches the temperature setpoint, the IDLE LED extinguishes.

To turn the unit off, place the POWER Switch to the OFF position.

When the unit is shut off, wait approximately five minutes before restarting. This allows time for the refrigeration pressures to equalize. If the pressures are not allowed to equalize, the compressor will short-cycle (clicking sound) and no cooling will occur.



Digital Temperature Controller

Digital Controller

The digital controller controls temperature using a PID (Proportional-Integral-Derivative) algorithm. It is designed with self-diagnostic features and easy to use operator interface.

NEXT ENTER

Use this key to scroll forward through the menus and also to accept and save changes.

YES, ▲

This dual purpose key is used to answer Yes to Y/N questions or to increment numerical values upward for setting numeric values.

NO, ▼

This dual purpose key is used to answer No to Y/N questions or to decrement numerical values downward for setting numeric values.

The controller will not allow you to enter a value above the maximum or below the minimum value, or any illegal value. If you try to enter an illegal value the display will revert to its original value when the last digit was entered.

Operator's Loop

When the controller is first powered up it performs a quick self-test and then enters the Operator's Loop, displaying the reservoir fluid temperature. Press the **NEXT ENTER** key to view the setpoint.

SP displays the controller setpoint. The display will flash between SP and the actual setpoint number.

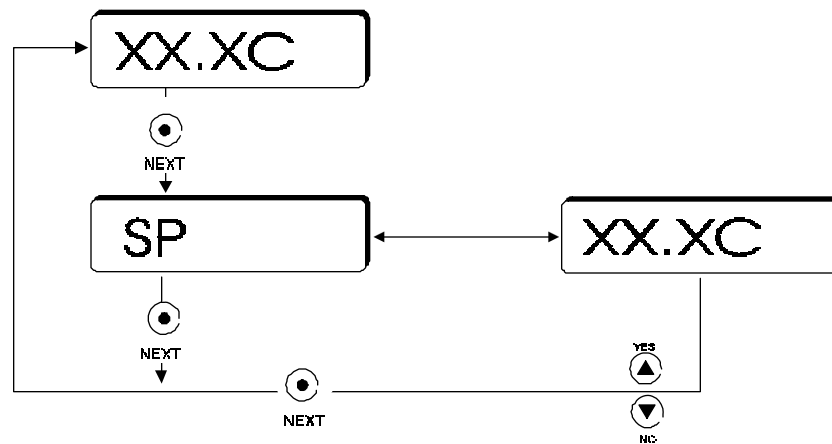
Ensure the setpoint is 20°C.

Use the **YES/NO** keys to change the setpoint value. Once the desired setpoint is displayed, press the **NEXT ENTER** key.

The new value will not be used by the controller until the **NEXT ENTER** key is depressed and the display stops flashing.

NOTE: If the **NEXT ENTER** key is not depressed within one minute, the controller will time out and the new value will not be accepted. The controller will revert to the previous value.

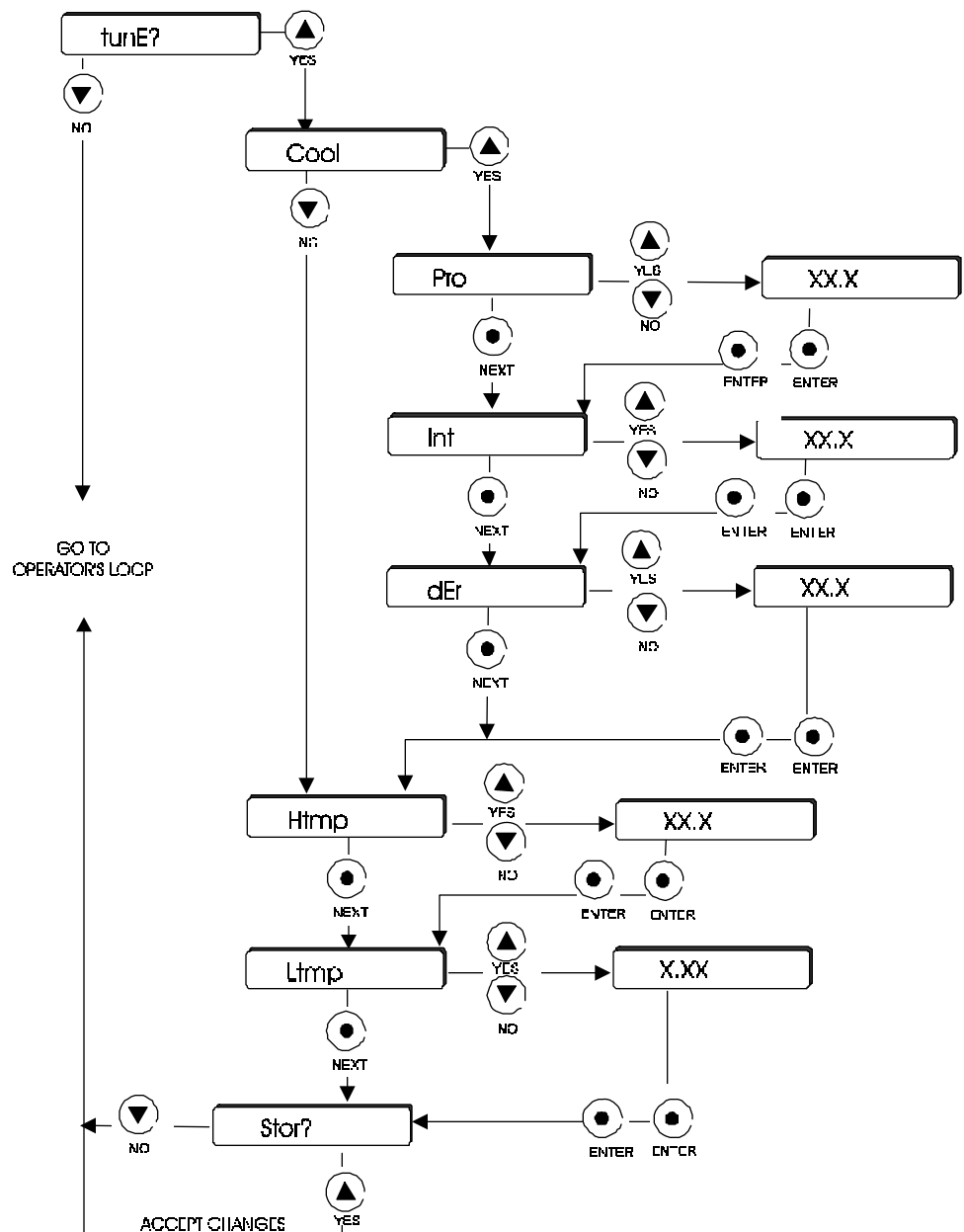
The controller will not allow you to enter a value above the unit's maximum temperature range or below the minimum temperature range. Trying to enter an illegal value outside the range, the display will revert to its original value.



Setup Loop

The Setup Loop is used to adjust the controller's PID parameters (Pro = Proportional, Int = Integral, dEr = Derivative) and adjust the temperature alarm limits (Htmp/Ltmp). The display will indicate tunE?

Enter the Setup Loop from the Operator's Loop by pressing and holding the **NO** key then pressing the **NEXT ENTER** key. The display will indicate tunE? Use the YES/NO keys to adjust the values. Press the **NEXT ENTER** key twice to accept the new value. **NOTE:** If the value was not changed, press the **NEXT ENTER** key only once.



NOTE: Should you desire to return to the temperature display and abort any changes, keep pressing the **NEXT** until the display reads **Stor**, then press **NO**.

Error Codes

The controller also has the capability to display error codes.

On Start Up:

These messages will be displayed on the controller. The controller will wait for someone to press the **NEXT ENTER** key. Once the **NEXT ENTER** key is pressed the controller will restore factory defaults and begin to run. For example if "Er01" is on the controller display at power up, this indicates the internal ram is corrupt.

Er01

Ram test failure.

Er02

Keypad failure, shorted keypad.

Er03 - Critical checksum failure, will restore all factory defaults.

Er04

Non-Critical checksum failure, will restore ½ of factory values, keeps calibration, PID, restore setpoint.

Unusual Hardware Conditions

These errors will flash on the display and cannot be cleared. These are internal controller problems.

Er04 through Er05 - Interrupt errors during runtime

Functional/ Machine errors

These are the runtime/ logic errors. These errors will clear themselves once the problem disappears.

Display	Indication
16	Bad calibration data
19	Temperature below low temperature setpoint
21	Temperature above high temperature setpoint
22	Excessive high temperature, unit requires immediate attention
23	Shorted RTD2
24	Open RTD2
25	Shorted RTD1
26	Open RTD1
31	Excessive fluid low level, unit requires immediate attention
ADD	Fluid low level

The unit will continue to run with any of the above codes. If any other code appears contact NESLAB customer service.

Displaying Software Version Number

The controller can display the installed software version number. For example, for a unit with software version **000507.44D**:

1. Unit is running normally and displaying reservoir fluid temperature.
2. Press and hold the **NO** key for at least 10 seconds. The display will show the digits to the left of the decimal, for example: **0507** (note the two leading zeros are not shown).
3. Press the **NEXT ENTER** key.
The display will show the digits to the right of the decimal, for example: **44**.
4. Press the **NEXT ENTER** key.
The display will show the revision letter, as its equivalent number, for example, **D = 4**.
5. Press the **NEXT ENTER** key.
Disregard this display.
6. Press the **NEXT ENTER** key.
The display returns to the reservoir fluid temperature.

Cleaning

Reservoir

Periodically inspect the fluid inside the reservoir. If cleaning is necessary, flush the reservoir with a cleaning fluid compatible with the circulating system and the cooling fluid.

The cooling fluid should be replaced periodically. When operating at low temperatures, the concentration of water in the cooling fluid will increase over time, leading to a loss of cooling capacity.

Before changing the cooling fluid, raise the operating temperature of the unit to deice the cooling coils. Refer to Filling Requirements on page 9 for instructions on replacing the cooling fluid.

Condenser

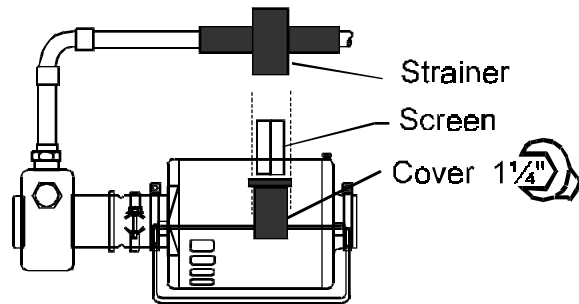
For proper operation, the unit needs to pull substantial amounts of air through a condenser. A build up of dust or debris on the fins of the condenser will lead to a loss of cooling capacity.

The lower front of the unit has a one-piece grille assembly. Gently pry the assembly off with a flathead screwdriver. Use care not to scratch the paint.

Periodic vacuuming of the condenser fins is necessary. The frequency of cleaning depends on the operating environment. We recommend a visual inspection of the condenser be made monthly after initial installation. After several months, the frequency of cleaning will be established.

Pump Strainer

Units have a pump strainer. If debris is in the system, the strainer prevents the material from being drawn into the pump and damaging the pump vanes.



After initial installation, the strainer may become clogged with debris and scale. Therefore, the strainer must be cleaned after the first week of installation. After this first cleaning, a biannual visual inspection is recommended.

Before cleaning the strainer, disconnect the power cord from the power source and place some absorbent material below the strainer assembly in case there is slight spillage.

Remove the wrapper from the unit. The strainer is located on the pump suction line. Place a cloth or sponge under the strainer cover to absorb any excess water. Remove the 1 1/4" cover to access the strainer.

Use a toothbrush and clean the screen by rinsing it with water.

When the screen is clean, replace it in the strainer, tighten the cover and replace the wrapper. Refer to Filling Requirements on page 10 for instructions on replacing the coolant.

Leaks

The flange area of the pump has assembly holes at the top and the bottom. Fluid leaking out of either hole indicates a shaft seal failure. The seal has a limited ability to seat around imperfections. If this is a new pump, allow it to run for 24 hours and reinspect. A continuous leak will require pump replacement.

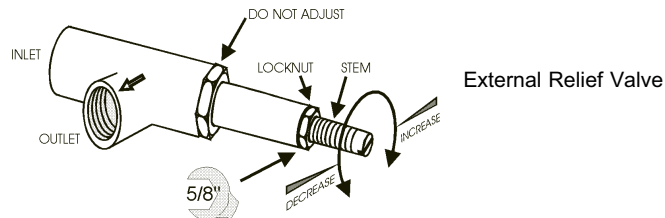
Certain fluids will deteriorate the pump shaft seal. Highly deionized water will leach ions out of the seal material, causing it to lose flexibility. This will cause leaks. Deionized water must be used with caution, and with monitoring of the deionization level.

Pressure Relief Valves

NOTE: The unit's pump is factory preset not to exceed 60 psi.

The unit has two relief valves: An internal one in the pump, and an external one on the pump discharge line. Because they interact, adjustment must be done in a certain sequence.

Turn unit off. Remove the wrapper and locate the external relief valve. Loosen the locknut. Turn the threaded stem fully clockwise.

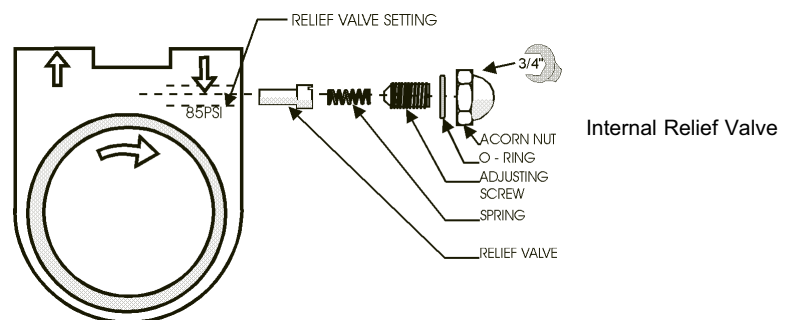


If the unit is not plumbed to an application, install a loop of hose between the supply and return fittings. Turn unit on.

NOTE: The external relief valve may drip while the locknut is loose.

Crimp hose flat using a pair of vise-grips. Use an external pressure gauge to indicate the internal relief valve setting.

Remove the acorn nut on the side of the pump. Adjust the pump pressure adjustment screw (under the acorn nut) until the gauge indicates the value stamped on the pump. (If not shown, adjust to 90 psi (620 kPa).) Turning the screw clockwise increases the pressure setting. Replace the acorn nut.



Back out the threaded stem on the external relief valve counterclockwise. As the setting drops below the internal valve setting, the gauge indicates the external valve setting. Continue until the gauge indicates 80 psi (550 kPa) or the desired setting. Tighten the lock nut.

The external relief valve is intended for coarse pressure limiting, not regulation. For precise pressure regulation, use an external bypass type regulator.

NOTE: The minimum pressure achievable using the provided external relief valve is approximately 35 - 45 psi (240 - 310 kPa).

Algae

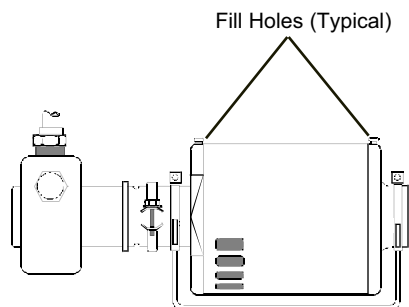
To restrict the growth of algae in the reservoir, it is recommended that the reservoir cover be kept in place and that all circulation lines be opaque. This will eliminate the entrance of light which is required for the growth of most common algae.

NESLAB recommends the use of Chloramine-T, 1 gram per 3.5 liters. Other algicides can be harmful to the unit's internal components. Contact NESLAB for additional information.

Pump Motor Lubrication

PD pumps use sleeve type bearings with large reservoirs. Oiling instructions are generally posted on each motor. In the absence of instructions, add approximately 30 to 35 drops of SAE 20 non-detergent oil to each bearing on the following schedule (SAE 20 = 142 CS viscosity):

Duty Cycle	Oiling Frequency
Continuous	Once every year
Intermittent	Once every 2 years
Occasional	Once every 5 years



Hoses

The unit's internal and external hoses and clamps should be inspected and tightened on at least a semiannual basis.

Refrigeration Data

Refrigerant	R-22
Amount	29 ounces (822 grams)
Hot gas setting	25 psig (172 kPa)
Ambient	68°F
Fluid temperature	20°C
Suction pressure	70 psig (480 kPa)
Discharge pressure	270 psig (1860 kPa)
Speed check	2.8°C/minute

Spare Parts List

Refer to illustrations on next page.

Line item	GE Part#	FRU Code	Description	NESLAB Part#
1	2222564-4	2	Nylon Coupling	004158
2	2222564-5	2	Water Pump/Nylon Coupling	008575
3	2222564-6	2	50/60Hz Pump Motor	008126
4	2222564-7	2	Remote Temp Device	005720
5	2222564-8	1	.25A/250V Fuse	010639
6	2222564-9	2	Controller	025413.2A
7	2222564-10	2	Pressure Gauge	025716
8	2222564-11	2	Strainer	004626
9	2222564-12	2	Relief Valve	009058
10	2222564-13	2	POWER ON/OFF Switch	006258
11	2222564-14	2	Transformer	025987
12	2222564-15	2	Solenoid Coil	009954
13	2222564-16	2	Castor	013686
14	2222564-17	2	Castor/Brake	013687
15	2222564-18	2	60Hz Fan Motor	007744
16	2222564-19	2	50Hz Fan Motor	007741
17	2222564-20	2	½" Nipple Hose Barb Water Fitting	012638
18	2222564-2	1	Water Chiller 60Hz	337104041505
19	2222564-3	1	Water Chiller 50Hz	337106041506
20	2223355	2	Power Cord	Supplied by GE
21	2138791	2	Antifreeze Coolant	Supplied by GE

Factory Preset Values

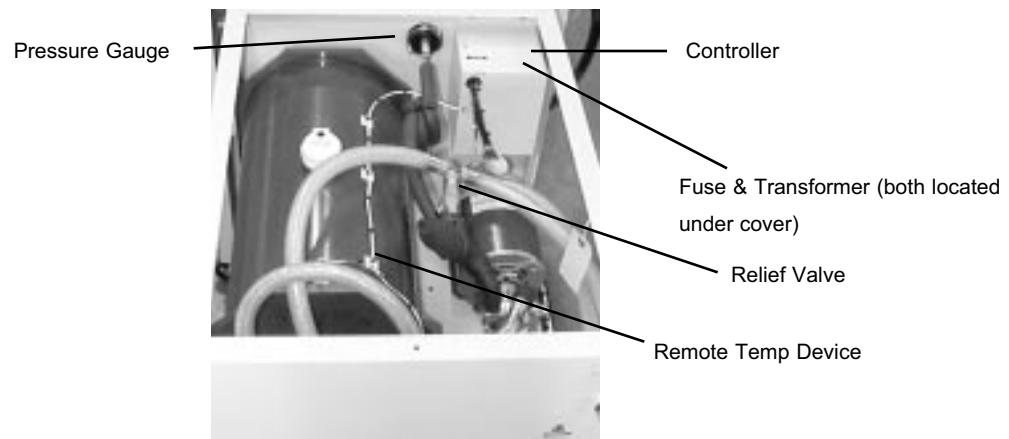
The following values are factory preset in the controller's Setup Loop.

PID Values

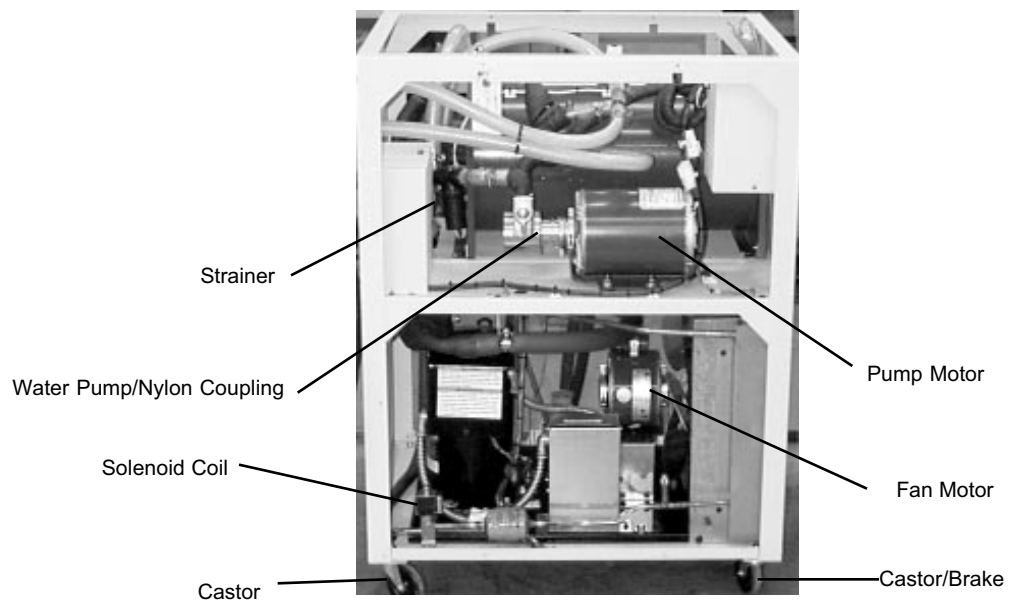
P	=	20%
I	=	0.5 repeats per minute
D	=	0.0 minutes

Temperature Limits

High Limit	+90°C
Low Limit	-20°C



Top View



Side View

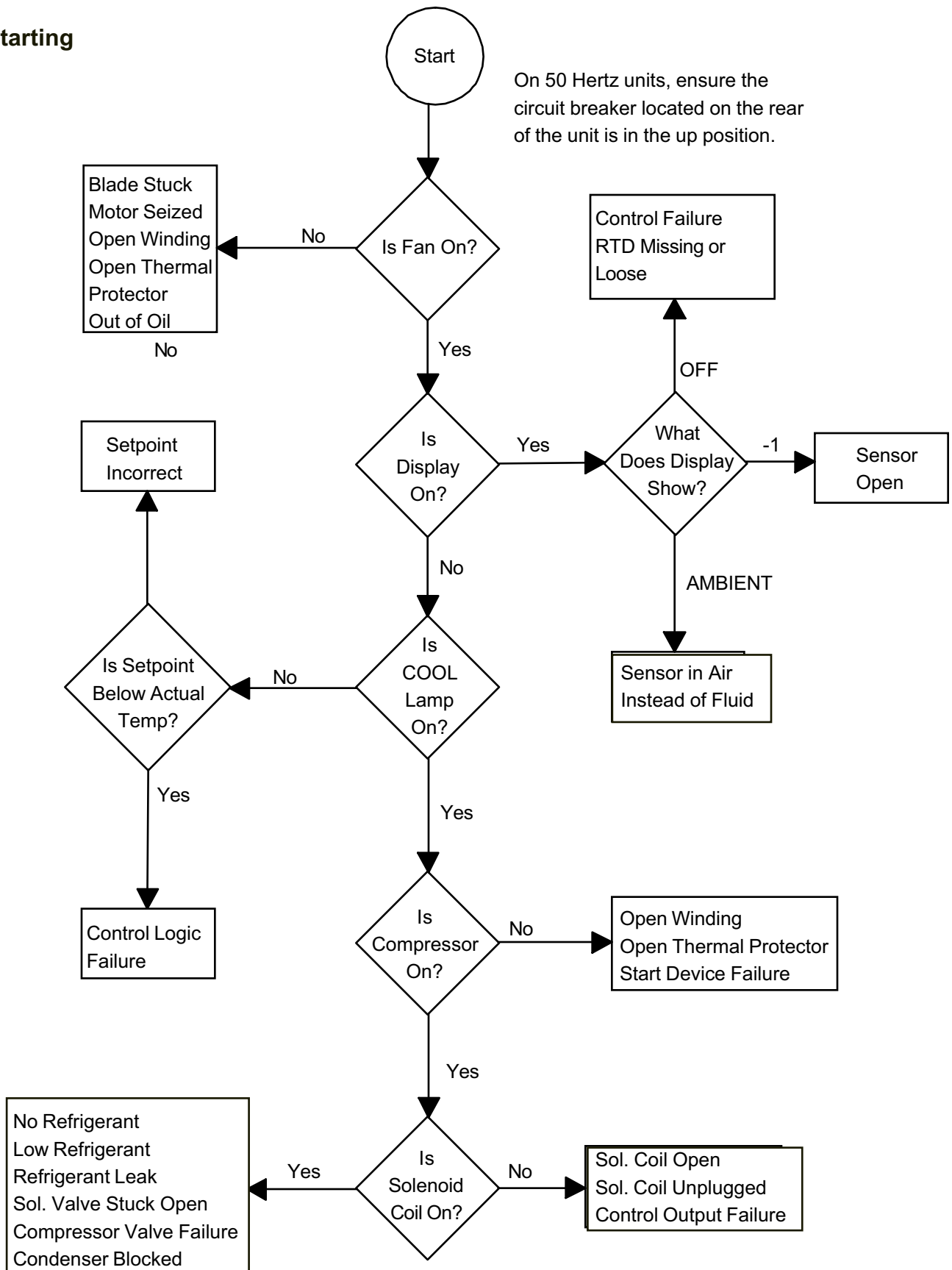
Section VI Troubleshooting

General

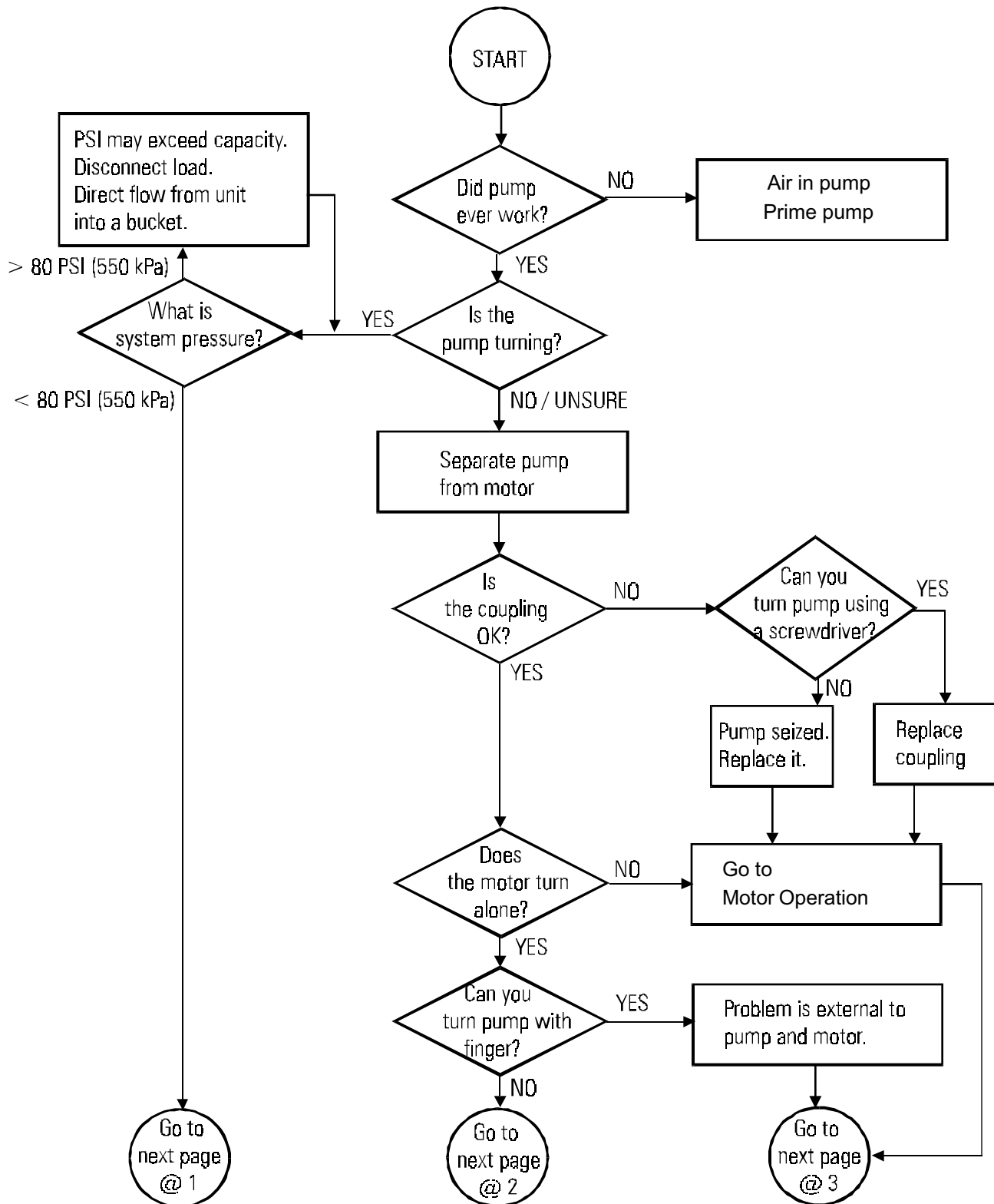
SYMPTOM	POSSIBLE CAUSE	CORRECTION
Circuit breaker trips	Compressor piston seized Compressor winding grounded Compressor winding shorted	Replace chiller Replace chiller Replace chiller
Control board triac opens	Solenoid coil shorted	Replace coil
Control dark but unit runs	Connector RTD loose/missing Controller failure	Restore RTD Replace controller
Control stuck in COOL	Sensor shorted	Replace sensor
Control stuck in HEAT	Sensor open or unplugged	Replace sensor or restore plug
Controller displays ambient temp	Sensor in air, not fluid	Top off fluid level
Controlling at wrong temp	Control out of calibration Sensor out of range	Recalibrate to temp standard Replace sensor
Fluid too cold	Controller output shorted / leaky Sensor in air, not fluid Solenoid valve stuck open	Replace controller Top off fluid in tank Replace valve
Fluid too hot	Refer to flowchart	
Frost on compressor inlet line	Normal if > 1" away from shell	Insulate if a concern
Frost on compressor shell	Too much refrigerant	Adjust charge
Fuse blows in supply circuit	Compressor piston seized Compressor winding grounded Compressor winding shorted	Replace chiller Replace chiller Replace chiller
GFCI trips in supply circuit	Electrical fault to ground	Isolate individual components

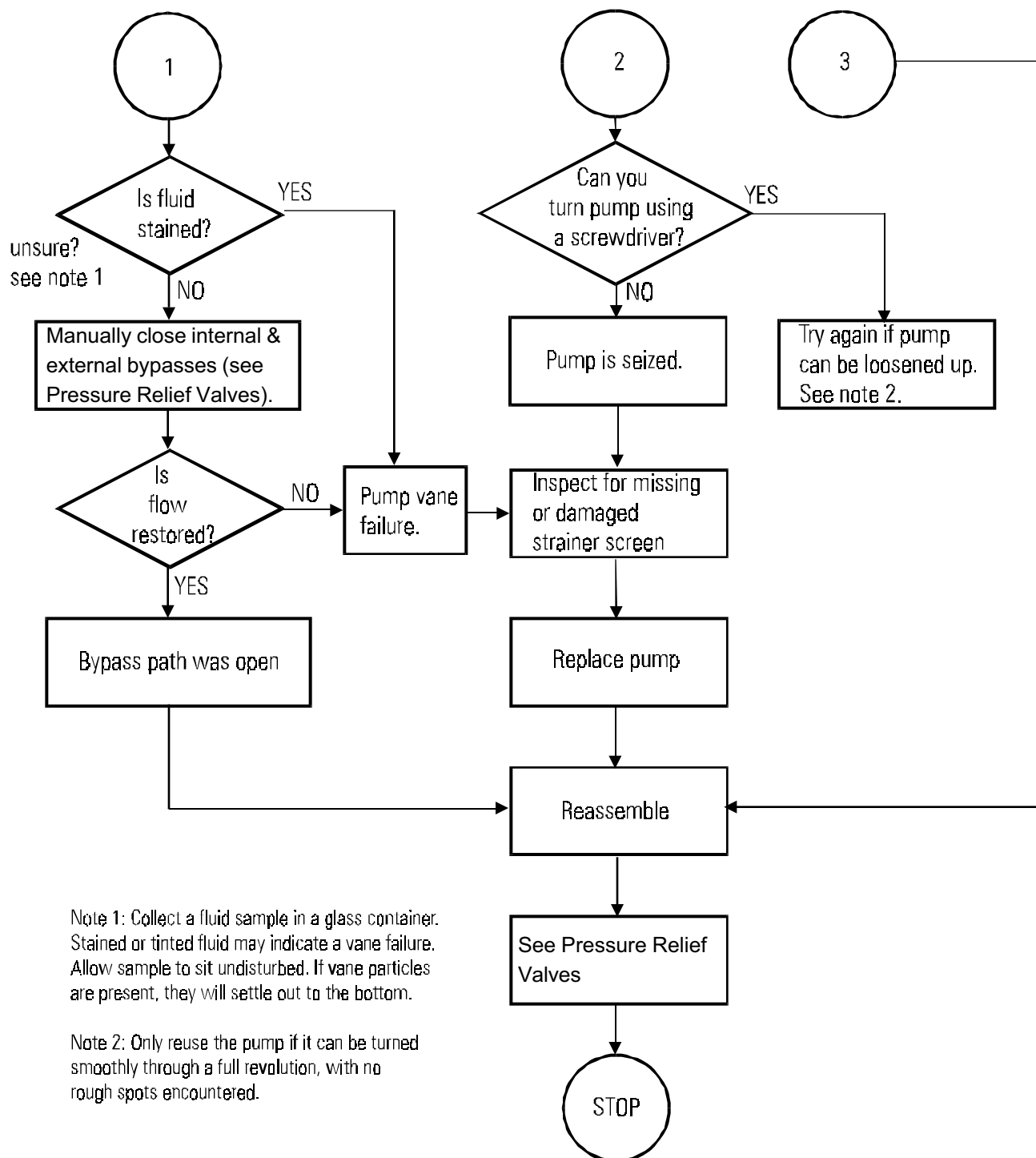
SYMPTOM	POSSIBLE CAUSE	CORRECTION
No flow but pump motor running	Pump impeller decoupled Pump not primed Valve closed external to unit	Clean/replace impeller Open discharge line to prime Open valve
No flow, pump motor off	Pump motor too hot Pump motor winding open	Allow to cool Replace motor
No start, but circuit live	Transformer open	Replace transformer
Noise	Compressor valve wear Debris in pump Fan motor bearings dry or worn Fan motor overloaded Pump motor bearings worn Pump motor bearings dry	Replace chiller Open and clean impeller area Oil wick or replace motor Clean condenser Replace pump motor Oil with #20 SAE elec. motor oil

Starting

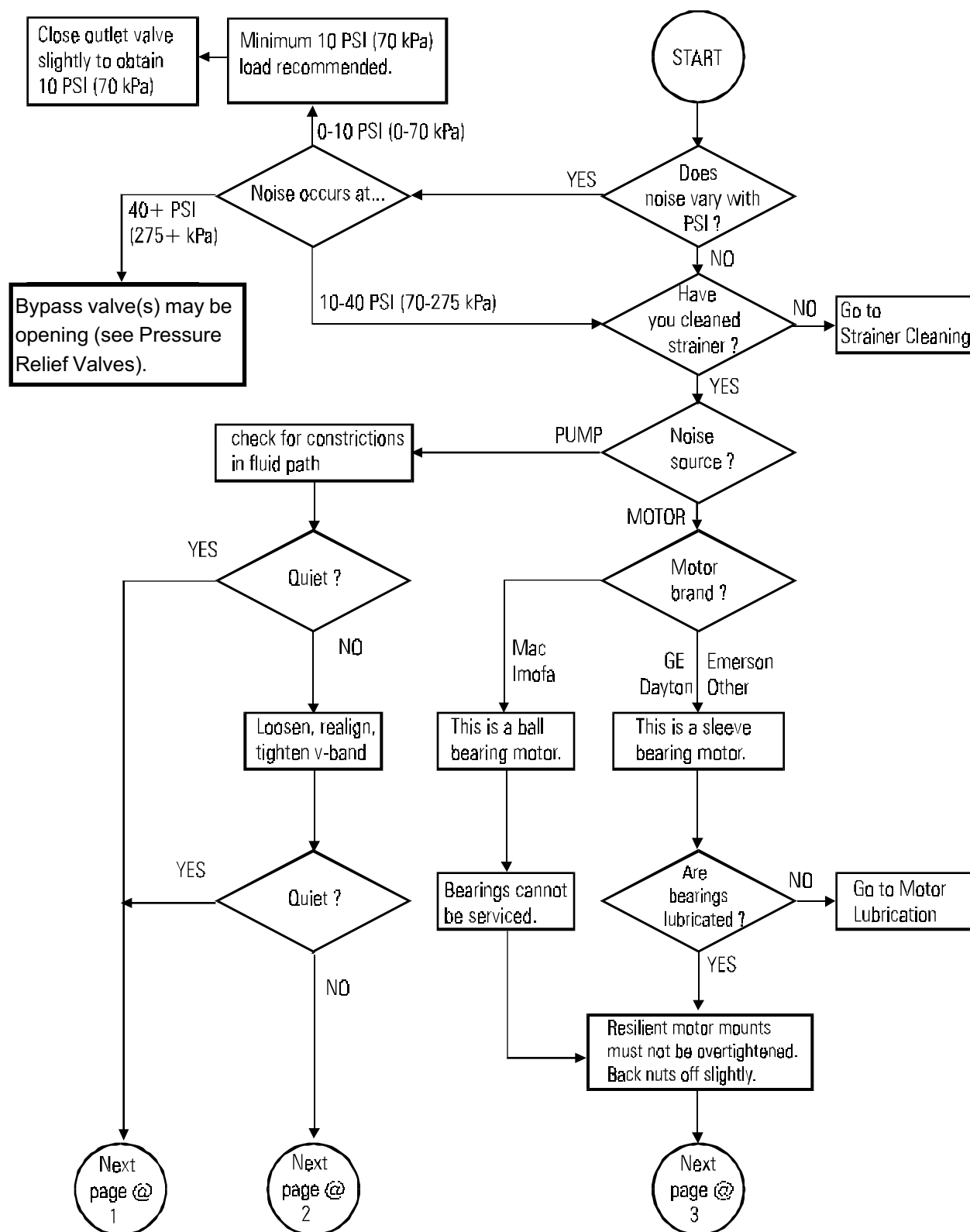


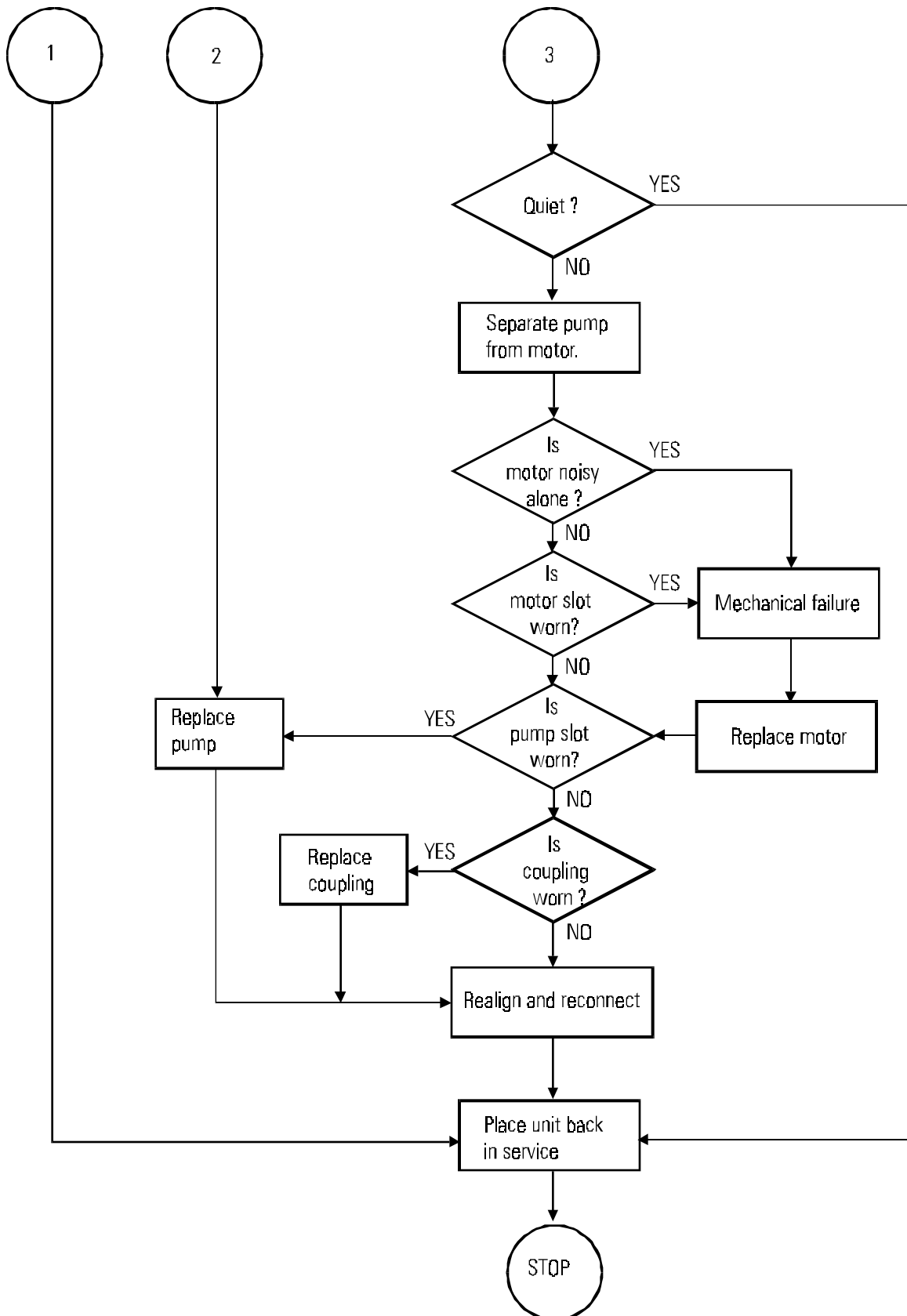
No Flow



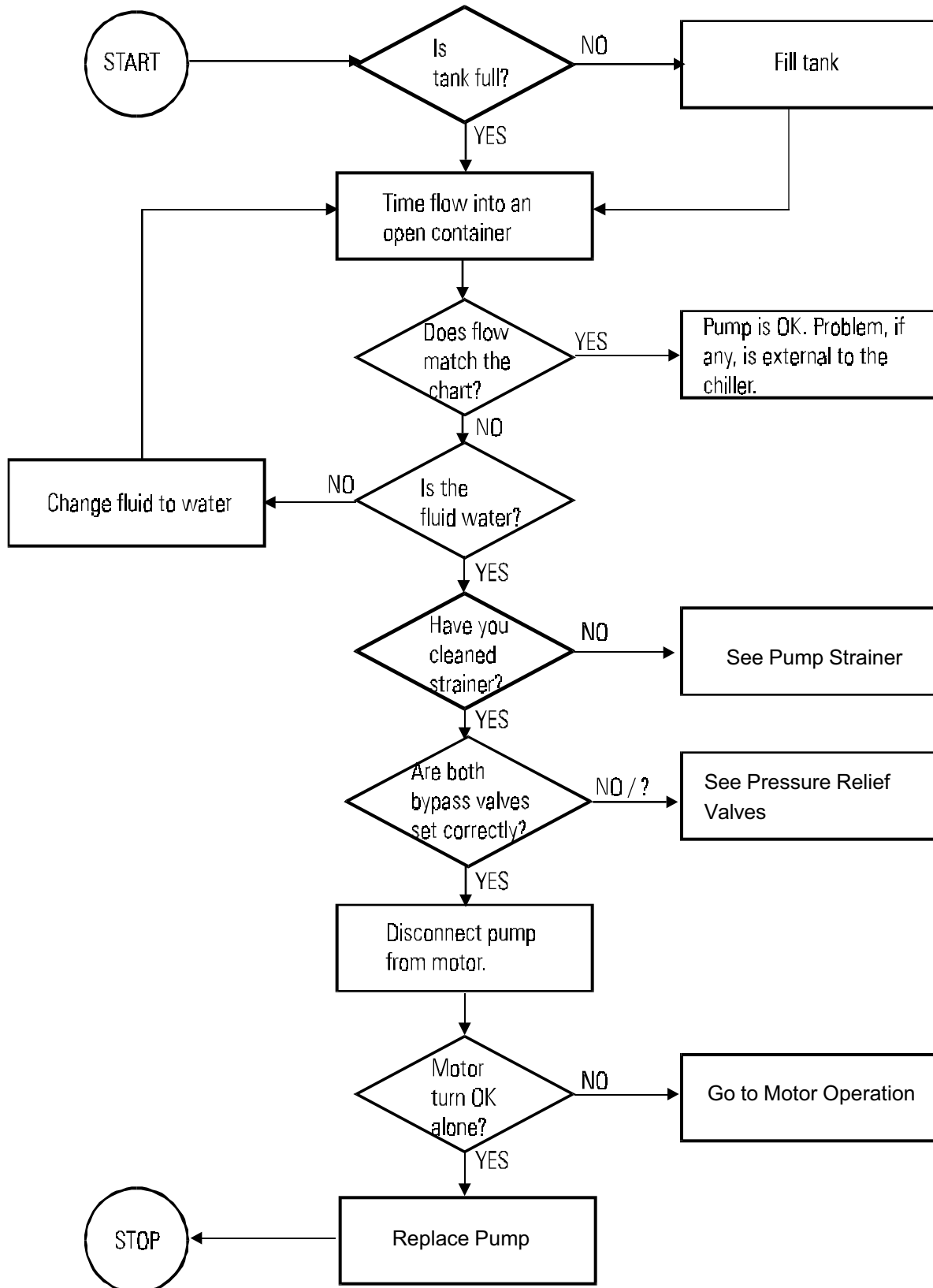


Pump Noise

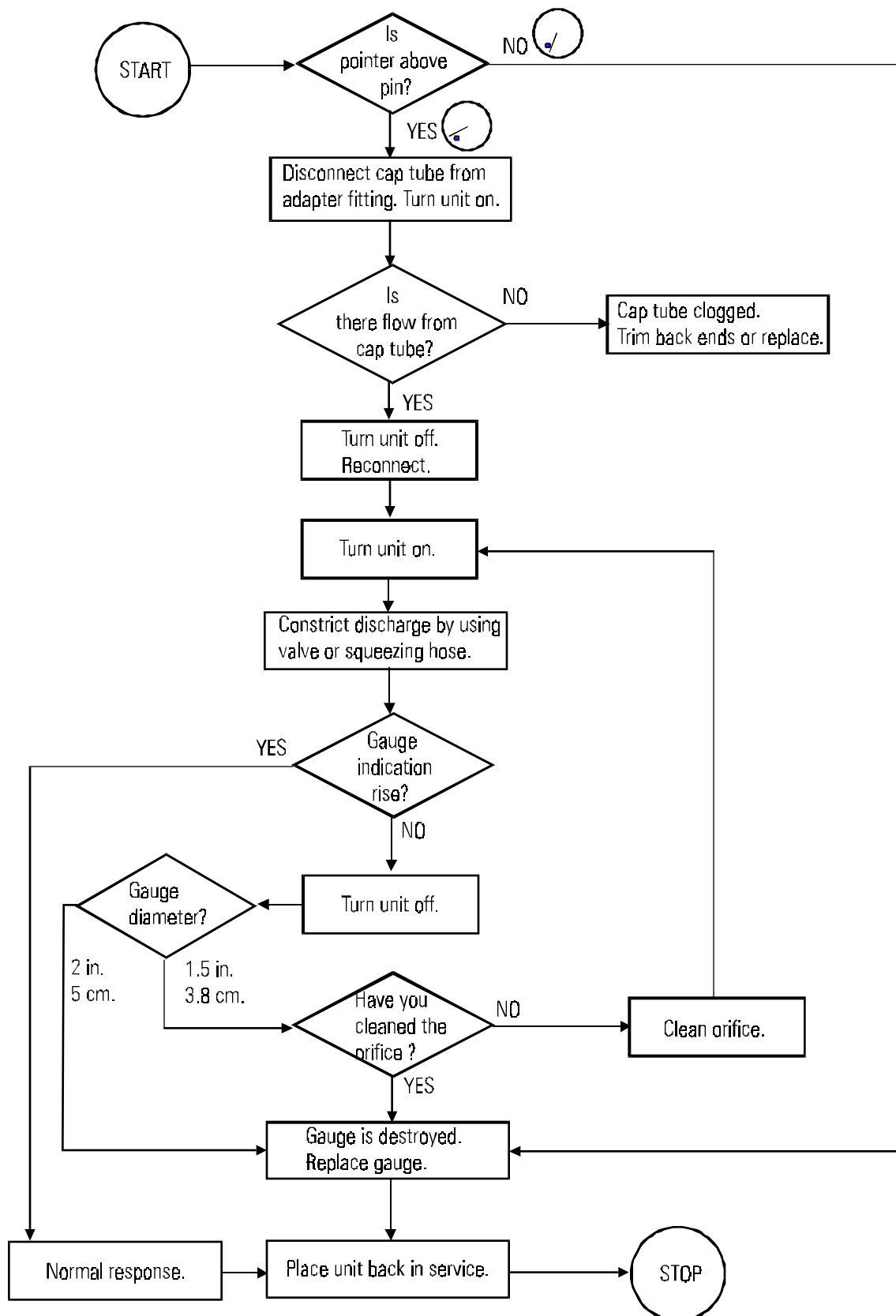




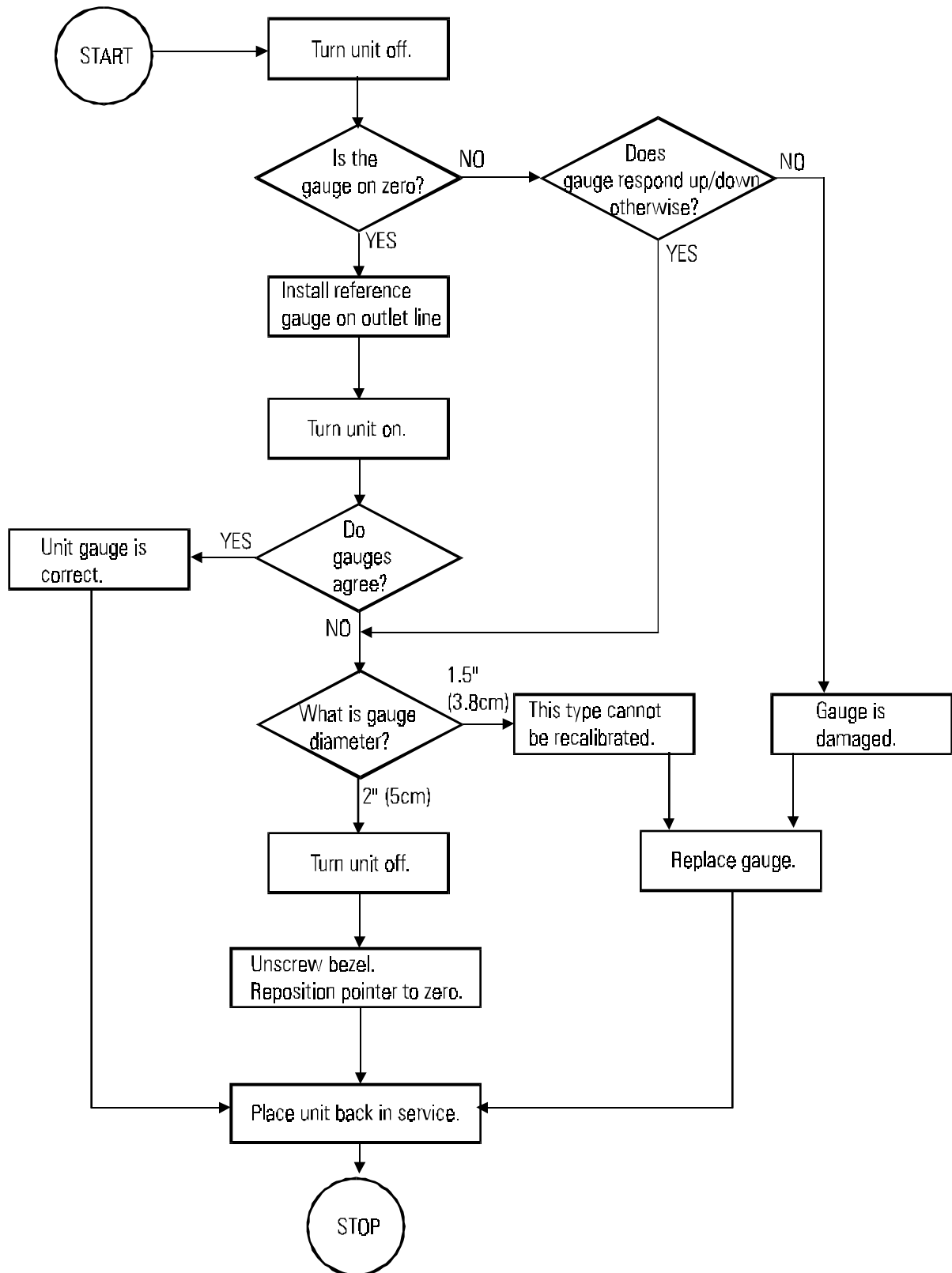
Low Flow and/or Pressure



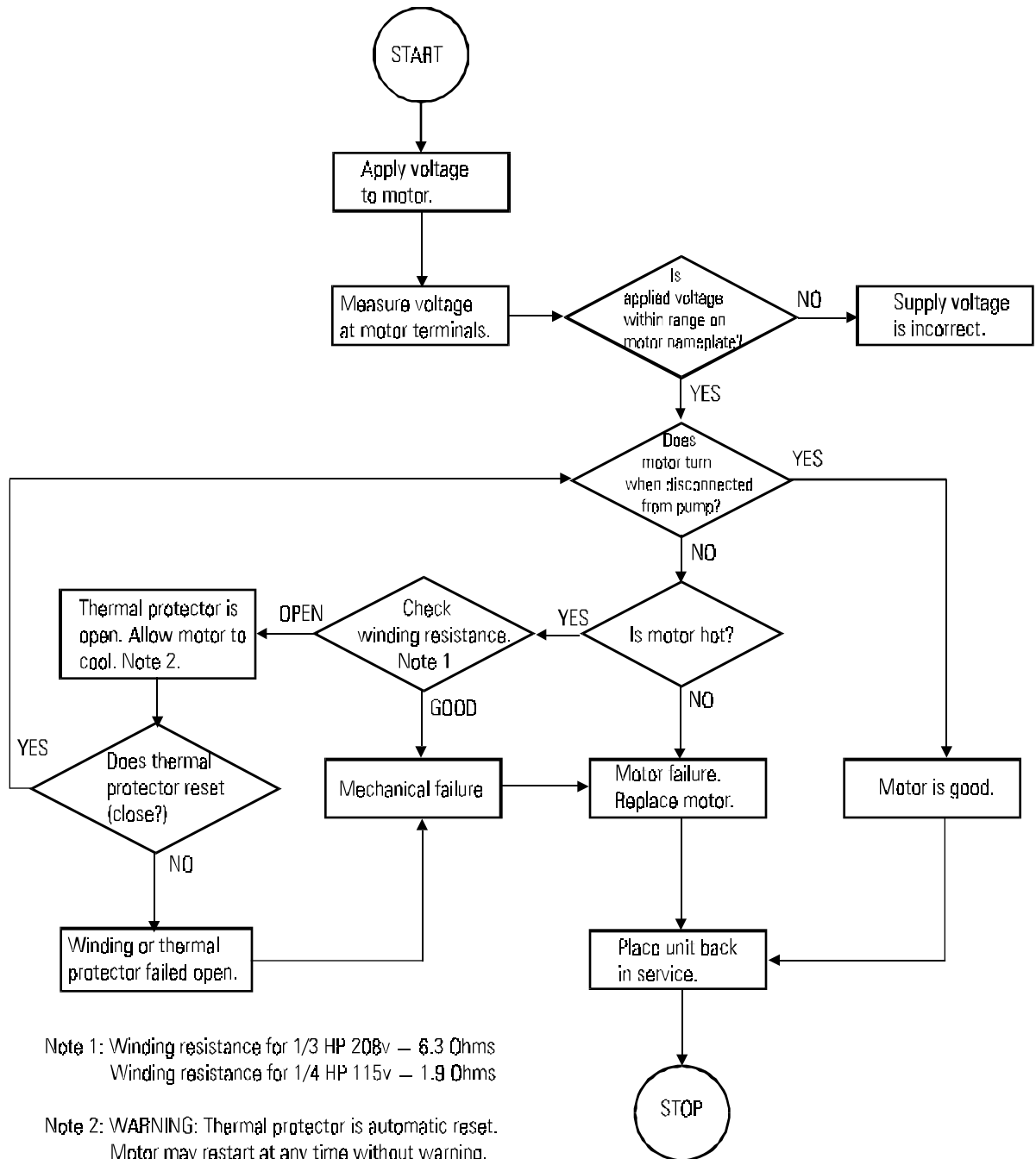
Pressure gauge always zero



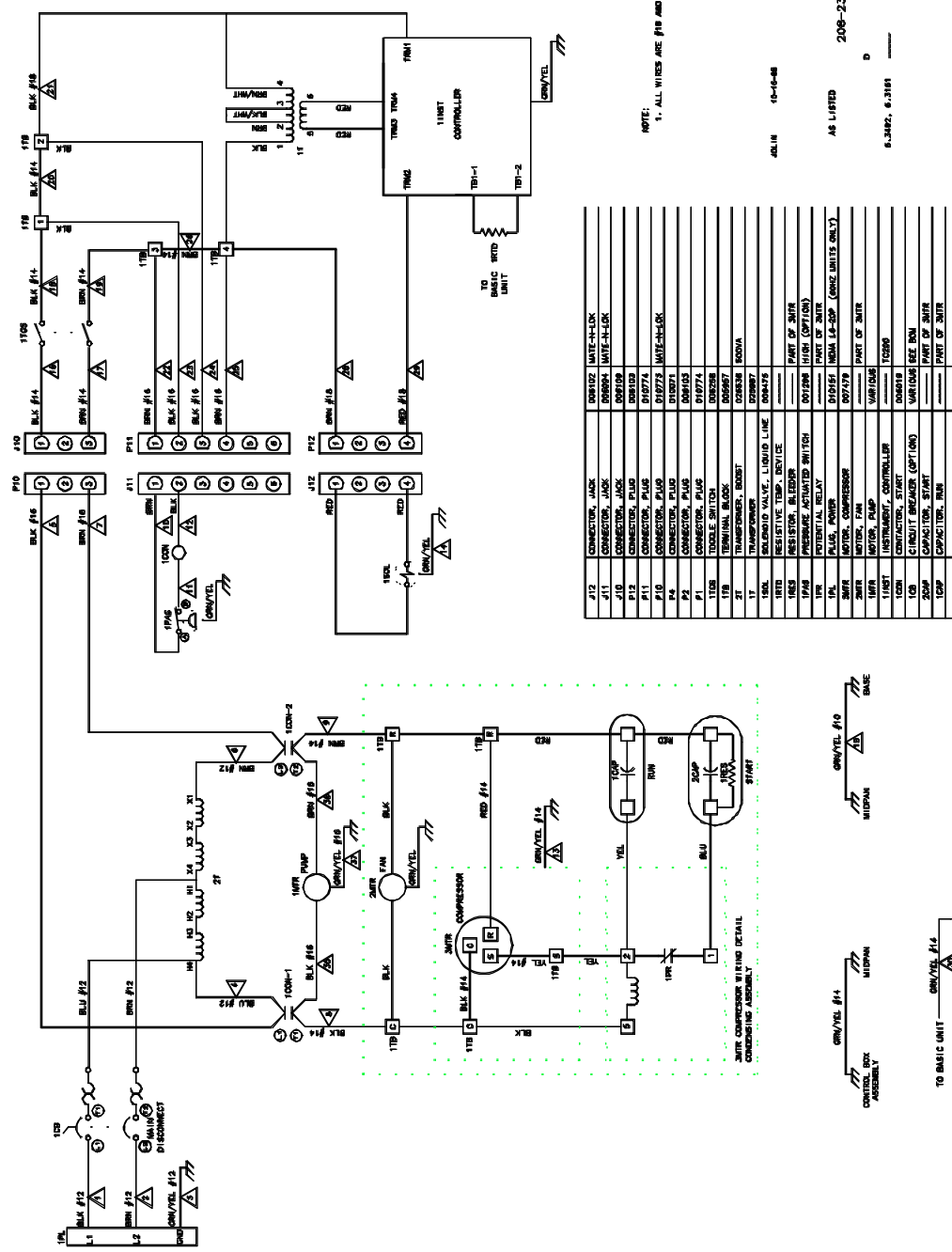
Pressure gauge incorrect



Motor Operation



Wiring Diagram (50 Hertz units)



NOTE:
1. ALL WIRES ARE #18 AND UNLESS OTHERWISE SPECIFIED.

CFT-150, CE97
DIGITAL UNIT
AS LISTED 200-230V, 50/60HZ, 1PH, R22
6.3850
D
6.3852, 6.3181
1 OF 1